W05 DRINKING WATER QUALITY MANAGEMENT | O (MAX : 3 PT)

Intent : Maintain and display consistent high quality of drinking water.

Summary : This WELL feature requires pre-testing of water quality parameters to determine treatment needs, monitoring at a more frequent interval and disclosure of water results.

Issue : Providing potable water to buildings is a multi-stage process that involves sourcing of the water, potabilization in treatment plants, distribution through network of pipes and delivery to the tap. While steady delivery of potable water is a reality in many places, other places must regularly contend with water that is delivered below potability standards or with fluctuating quality due to the intrusion of contaminants in the water distribution pipes,[1] unsupervised changes in municipal water supply and treatment[2] or weather-related events.

Solutions : From a building perspective, sound water quality management begins with an understanding of the incoming water quality, preferably through testing and analysis of historical data. If needed, treatment devices, such as filters or UV disinfection units, can be used to achieve data-driven, health-based water quality targets.[3] Periodic water monitoring not only confirms the quality of the water, but also helps to determine the needs for maintenance in pipes, fixtures or treatment devices. Availability of water quality results and maintenance records to occupants may also increase drinking water consumption, furthering both cost-saving and sustainability efforts for the project, while also promoting occupant hydration.

PART 1 ASSESS AND MAINTAIN DRINKING WATER QUALITY (MAX : 2 PT)

For All Spaces:

1: Water quality pre-test

For first-time registered projects, the following requirements are met:

a. The project pre-tests water at least one month before Performance Verification for the parameters below:
   1. Turbidity.
   2. Coliforms.
   3. pH.
   4. Total Dissolved Solids (TDS).
   5. Total Chlorine.
   6. Residual (free) chlorine.
   7. Arsenic.
   8. Lead.
   9. Copper.
   10. Nitrate
   11. Benzene.

b. Sampling occurs at the following locations:
   1. The water dispenser that is closest to the pipe that delivers water into the project, before any point-of-entry water treatment system where possible.
   2. For projects with more than two floors or more than 930 m^2 in area, a second drinking water dispenser on the highest floor to which the project has access that is farthest from the location in requirement b(1).
   3. For projects of 12 or more floors, one additional drinking water dispenser for every 10 floors.

c. Samples must be taken with point-of-use filters or other water treatment devices bypassed or removed, if present.

Note : Projects pursuing re-certification do not need to achieve Option 1.

2: Water quality monitoring

The following requirements are met:

a. Piped water is delivered to drinking water dispensers.

b. Water is tested quarterly in drinking water dispensers and meets the following thresholds. If any sample exceeds these thresholds, remediation and re-testing occur within a month:
   1. Turbidity is 1.0 NTU, FTU or FNU or less.
   2. pH is between 6.5 and 9.0 (between 5.5 and 9 if a reverse osmosis system is installed at the point of use).
   3. Total Dissolved Solids (TDS) are 500 mg/L or less.
   4. Total Chlorine is 5 mg/L or less.
   5. Residual (free) Chlorine is 5 mg/L or less.
   6. Total Coliforms are not detected in a 100 ml sample. Testing is required only if residual chlorine is not detected.
   7. Lead is 10 µg/L or less. Sampling frequency can be reduced to once per year if results are below detection limits in two consecutive samples.
   8. Copper is 2 mg/L or less. Sampling frequency can be reduced to twice a year if results are below 0.1 mg/L in two consecutive samples; testing is no longer required if four consecutive samples are below this threshold.

c. The number and location of sampling points for on-going monitoring complies with the requirements outlined in the Performance Verification Guidebook. For pH, use sampling locations and frequency set for residual chlorine.

d. All test results are submitted annually through the WELL digital platform.

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 PROMOTE DRINKING WATER TRANSPARENCY (MAX : 1 PT)

For All Spaces:

The following information is prominently displayed near sources of drinking water (or on a website available to occupants):

a. Water quality results from the most recent sampling, including date of testing and compliance with WELL thresholds.
b. If filters or other treatment units are in use, information about the treatment technologies and most recent date of device maintenance and/or filter cartridge replacement.

REFERENCES

1. World Health Organization. Water Safety in Buildings. Geneva, 2011.
2. Torrice M. How Lead Ended Up In Flint's Water. C&EN Global Enterprise. 2016;94(7):26-29.
3. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.